

Calmbots: Exploring Possibilities of Multiple Insects with On-hand Devices and Flexible Controls as Creation Interfaces

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Figure 1: We explore multiple insects as co-creation interfaces towards art works in daily life. E.g., drawing using calligraphy pen (center).

ABSTRACT

We introduce "Calmbots" that are insects-based interfaces comprising multiple functions, i.e., drawing, display, transportation, or haptics. We explore possibilities of multiple insects as co-creation interfaces making artworks through insects indirectly, in contrast with BioArt where artworks are made by biological technology. Considering scalability, sustainability, or promotability of applying insects-based devices on creation activities in daily life situations, we utilize on-hand and efficient control system by AR markers and radio-based station, and propose efficient ways of controlling multiple insects reaching goals and transporting objects, and customize flexible option parts. In robust experimental trials, our results showed effective control on a group of three or five cockroaches with at-least 60% reaching accuracy, mobility on carpeted or cable-lines floor, and possible continuous control under certain time duration. Participants in a user study felt positive with Calmbots' functions and expected more performance or appearance on activities or creation in daily life.

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CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Interfaces of Multiple Insects, Insect-based Creation

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1 INTRODUCTION

There have been multiple researches on design, functions, or applications on multiple and diverse insects or creatures [8, 13, 32, 49, 58, 69]. In art creation, there are rich artworks that are made by biological technologies or insects or creatures themselves [5, 15, 16, 22, 26, 43, 52, 59, 60, 64, 65]. However, few studies had been discussed about applying the tools of insects or creature to co-create artworks indirectly, which might be interesting especially in the era of post COVID-19. In this study, we aim to explore and design co-creation interfaces through insects' capabilities, for assisting users or artists in artworks or creation activities. One drawback of applying common robotics technology in art co-creation is that

mechanical robotics may be limited under some conditions, such as on vertical, narrow, or uneven floors owing to their limited mobility. Another rationales we consider applying insect artists are related to the concepts of *ubiquitous computing* and *calm technology* [70]. For example, according to Mark Weiser, “*The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.*” [70].

We consider to utilize insects’ capabilities to build interfaces that can be flexibly and conveniently used by human. The possibility of ubiquitous interfaces and applications by insects may not be limited to specific spaces, timing, or functions, compared with existing many Tangible User Interfaces (TUIs) [51]. In building the interface, we get inspiration from cockroaches, specifically, the Madagascar hissing cockroach¹. Owing to their simple nervous system, this species of cockroaches can be easily controlled via electrical stimulation. Further, they can move on uneven floors and exhibit remarkable strength; for instance, they can move common lightweight objects such as tablespoons, and co-exist in human living space smoothly. Additionally, they can be designed to be working or non-working and are capable of self-maintenance as insects, which are different with actuator-based robots.

In co-creation interfaces through insects and considering scalability, sustainability, or promotability of applying insects-based interfaces in daily life creation activities, we design Calmbots, which are composed of multiple cockroach robots with defined functions in living spaces, including object transportation, display, drawing, haptics, sensing, and sound. First, we perform surgery on Madagascar hissing cockroaches to make them controllable via electrical stimulation. We also attach customized optional parts to these cockroaches depending on the intended purpose and built a system that can track and communicate with these backpacks in real-time applications using augmented reality (AR) markers and a radio-based station. To achieve the necessary functions in living spaces, we propose the processing steps for controlling multiple cockroaches to reach their goals, transporting objects, and drawing (e.g., letters). In addition, we propose optional parts with customizable shapes or materials to achieve object transportation and drawing.

In experimental results, we evaluated Calmbots based on experiments that included controlling multiple cockroaches to reach defined goals. We also investigated the effect of floor flatness on mobility, workload (continuous controlling success), the effect of stimulating body parts, and control performance across successive experiments with breaks. Calmbots exhibited effective goal attainment ability (87.5% and 60% accuracy for 3 and 5 cockroaches, respectively). Furthermore, cockroaches moved effectively on carpeted or cable-lined floors. Additionally, we ascertained that the individual cockroaches remained effective within a certain duration. We also observed that the performance of the individuals under electrical stimulation was not affected by previous experiments. With the abilities of maneuverability, mobility on uneven floors, and a certain level of ability of working continuously, Calmbots might be capable of executing multiple daily life tasks, including object transportation, by controlling multiple cockroaches.

In user studies, users showed positive feedbacks on Calmbots’ functions (transportation, drawing, or haptics) and expected interesting or usable tasks in living spaces or remote conditions, where some of them could be useful in co-creation activities, such as remote painting on the wall as graffiti, remote sensing or object transportation at place people hard to reach. On the other hand, some users concerned the quality of Calmbots performed on tasks required high details of contents. From experimental analysis and feedbacks of user study, we might infer as follows. Some users might be interested in co-creating simple and rough artworks indirectly or remotely through the tool of high movable and controllable insects, such as Calmbots. Additionally, for increasing quality of artworks, attachable option parts could be used to increase capability or creativity of insect-artists. Further, co-creation with Calmbots might be useful in simple visual arts, such as rough painting, printmaking, sculpture, or drawing on or with simple objects of rough shapes (rectangles, curves, spiral, or diagonals), especially in remote ways of co-creation activities.

2 RELATED WORK

2.1 Creation Activities through Creatures or Technologies

In bio-art creation, artists, practitioners, or researchers have utilized biological elements or technologies to create artworks, such as DNA artworks [5, 15, 16, 22, 64]. Recently, due to technology development, such as AI, artworks created by AI or intelligent robots has attracted high attention [11, 39, 46, 48, 50]. These kinds of art creation have raised diverse and controversial issues such as related ethics, studies, biologies, technologies, psychologies, aesthetics, or societies [26, 43, 52, 59, 60, 65]. In this study, we explore novel creation tools through insects, where users can co-create artworks by flexible control of insects’ movement and thus these artworks are made by insects indirectly. Therefore, human-insect co-creation may face challenges on capabilities of functions or attachment designed on insects, control quality of insects’ behaviors, and coordination between users and insect-artists. Further, usage perception or experiences of creation with insects and related appearance of insect tools may affect user engagement or motivation on co-creation activities.

2.2 Development of Remote Controlling Insects or Creatures

There are various ways to control the movement of creatures [35, 67]. Direct electrical stimulation is the most common and feasible approach [55] because it affords greater immediacy compared to other control methods. For example, in a study [21], the attempt was made to control a group of cockroaches by introducing a robot into the group. However, the results obtained were unsatisfactory. Therefore, in this study, we electrically stimulated multiple cockroaches to achieve immediate and accurate control. Control via electrical stimulation entails attempting to control movements by stimulating the nerves, muscles, and sensory organs of creatures. It has been studied for a wide variety of creatures, ranging from vertebrates, such as birds [6], rodents [69], fishes [8] and reptiles [57], to simple neural creatures such as insects [49] and even plants

¹Retrieved April 3, 2021, from https://en.wikipedia.org/wiki/Madagascar_hissing_cockroach

[32, 58]. Substantial research has yielded effective control of the movements of these creatures. Because the fundamental methods for controlling movements have been established, we aimed to utilize the creature-control technique in diverse applications, including in living spaces.

Multiple applications have exploited the advantages of immediate and accurate remote creatures control, considering creatures such as rodents and insects, using various control methods. *Latif et al.* [33] investigated how to maneuver a cockroach on a specified course using radio signals. In another study [55], cockroach control was automated using a tracking camera based on the research data of [33]. Further, various studies have been conducted in the context of search and rescue operations. An application has been developed that enables a cockroach to walk through a maze and remain on a specified course [34]. In another study, [73], a special cockroach attached to an ultrasound sensor was leveraged to effectively explore the surrounding environment. From the perspective of rescue operations using robots, terrain exploration using drones and robots that can be connected with the base control unit have been studied [13]. However, few studies exist on how to control multiple insects such as a group of cockroaches. Therefore, the proposed interface that leverages the advantages of multiple cockroaches (i.e., creature characteristics), such as mobility and hiding and self-maintenance abilities, for multiple and diverse applications in living spaces is novel. *Cyborg botany* [58] is the process of applying plant sensors in living spaces using special lab equipment; however, limited mobility and special equipment might limit the applications of plant sensors. Accordingly, our approach utilizes the advantages of using cockroaches (e.g., mobility) and controlling them using simple and low-cost surgical methods.

2.3 TUIs and Swarm Robots, Ubiquitous Robotics, and Limitations

Table 2 shows a comparison of TUIs, SwarmRobots and *Calmbots*. Most of the research on TUIs utilize stationary interfaces, which limit the maneuverability of objects, owing to the space restriction [25, 51]. Existing methods involve the use of pin arrays [45], electromagnet arrays [68, 71], ultrasound [47], electrostatic adsorption [3], and vibration [74]. Using small and movable robots, *Zooids* [19] constitute an interface with multiple functions, such as moving objects or providing haptic feedback. However, owing to the limitations of small robots (i.e., attached actuator), *Zooids* cannot easily move on nonflat surfaces (e.g., asphalt ground) and might be limited to flat and simple spaces. *Hermits* [44] contain multiple functions and can move on nonflat surfaces. However, they might be limited in moving on vertical planes or more complex grounds. On the other hand, *LiftTiles* [62] leverage pin-array devices that improved mobility and are usable in multiple living spaces. However, the large size of pins may limit users' interaction with digital information on a tiny scale (in centimeters).

Furthermore, there are problems pertaining to mobility or space in swarm robotics, which may be summarized as follows [53, 56]. In the human-computer interaction community, mouse-based maneuvering [30, 31] is used; the input is achieved through aerial gestures [2, 28], tablet devices [20], and through sketches [23]. Display information is set up as a physical display [1, 12, 36]. *Zooids*, *UbiSwarm*

(which exploited haptic capability based on *Zooids*) [29], *Rolling-Pixels* (a display device that moves by rolling) [37], and *ShapeBots* (which are designed to improve mobility through pins that can extend vertically and horizontally) [63], have implemented useful interfaces or functions for multiple situations, especially on flat or two-dimensional space. However, these studies may not be applicable under difficult conditions such as nonflat or asphalt surfaces and walls or ceilings that need to be climbed. In *GridDrones* [4], drones are used as an interface; however, although they are movable in three-dimensional space, they can only be deployed in large spaces. *FreeBOT* [38] robots can climb walls, demonstrating increased mobility owing to their use of magnets. However, they may not be applicable to common walls without embedded metal. Another problem in swarm robotics is that actuators must be attached to the robots. However, not only the actuators are noisy, but they also reduce the mobility of the robots. Therefore, we propose the use of the biological muscles of cockroaches as support for actuators. By using biological muscles as actuators, our cockroaches can maintain mobility while working quietly. We selected the Madagascar hissing cockroach, being an excellent climber and having strong legs and high mobility, to create powerful insect robots that can function in multiple daily life situations.

Ubiquitous robotics entails developing physical mobile platforms for ubiquitous computing such as multi robots, multiagent coordination, and localization. The focus in ubiquitous robotics has been on introducing mechanical robots into living spaces [27]. To this end, various models have been proposed from the engineering perspective, such as experimental multirobots [10, 42, 54], multiagent coordination [24, 41], and network localization [9, 40, 61]. In a study [14], an attempt was made to perform complex tasks in living spaces through the cooperation of multiple robots. However, these studies on ubiquitous robotics might not be applicable to the introduction of nonmechanical robots (e.g., insect robots) into living spaces because they might not support the method for controlling multiple insect robots for functions in living spaces as well as the required system and backpacks.

3 CALMBOTS DESIGN: CO-CREATION INTERFACES THROUGH INSECTS

3.1 Exploring Madagascar Hissing Cockroaches as Co-creation Interfaces

We present *Calmbots*, an exploratory interfaces of human-insects co-creation using cockroaches' capabilities. As we investigate appropriate creation activities, which can be co-created with insects' artists, we consider co-creation activities that are simple and relatively willing to be conducted by common users, especially novice users. With discussion and survey, visual arts, one of common artworks, may be target creation activities co-worked by human-insects since the required creation capabilities can be preliminary (e.g., drawing lines) and the generality of visual arts. Furthermore, recently, ubiquitous computing techniques have focused on methods of introducing flexible robots into living spaces to create smart living spaces [7]. Therefore, we consider applying insect co-creation interfaces in living spaces, and show the advantage of applying the species of cockroaches (*Madagascar hissing cockroach*) as insect artists, as follows.

- Easy maneuverability owing to their simple nervous system. We can easily control an individual Madagascar hissing cockroach through electrical stimulation because of the cockroach's clear response to electrical stimulation of its simple nervous system. Mobility. Madagascar hissing cockroaches usually have superior mobility; they can move between walls, uneven surfaces, glass surfaces, and sandboxes, unlike common insects.
- Strength. In living spaces, force is required to implement certain actions such as object transportation. The Madagascar hissing cockroach is strong and large and may be suitable for transporting heavy or large objects. Further, we conducted a simple experiment and observed that the maximum pulling force was 50 grams and the maximum pushing force was 30 grams. Consequently, the weight or size of the backpack (used for control) may not affect the mobility of the cockroaches.
- Hiding and self-maintenance abilities and noiselessness. Cockroaches can co-exist with humans in living spaces because they can hide; furthermore, they are capable of self-maintenance within a lifespan. Therefore, cockroach robots can be activated or deactivated, depending on the requirements of the human users. Note that the Madagascar cockroaches are noiseless, unlike common robots whose actuators are noisy.

With above properties, cockroach robots are possible to co-worked with users on multiple and diverse co-creation tasks, especially in visual arts and in living spaces.

3.2 Functions of Insects for Co-creation Activities

3.2.1 Object Transportation. We utilize cockroach robots for transporting objects, a function that is often required in daily life. Specifically, for transporting multiple objects of different sizes or weights, we design the necessary optional part (in Subsection 3.6), which is installed on the front of the cockroach, to which objects are attached. Thus, we could modify the shape of the optional parts, depending on the size of the object to be transported, and modify their shapes for enhanced grip when tightly attaching objects.

3.2.2 Display and Drawing for Symbols. We consider using cockroach robots for movable display or drawing, unlike common display devices. Specifically, we aim to control multiple individuals to display diverse information. For example, if the display information required by a user is the letter "A," we can control a few cockroaches to occupy each of the points comprising "A". Further, we control multiple cockroaches to convey useful information through drawing symbols, such as marks, at locations that are difficult to access (e.g., the ceiling of the room). Specifically, for this purpose, we attach a customized pen to the front of each cockroach and control multiple cockroaches to move, achieving the desired trail of a pen.

3.2.3 As Devices for Haptics, Sensing, or Sound. We propose the use of cockroaches as movable devices of haptics, sound, or sensing and execute tasks at the desired time because they can reach difficult-to-access spots (e.g., the ceiling). For example, we can control cockroaches to touch the body of objects, making sounds (additional

circuits are installed for triggering sound), or sensing the environment through attached sensors. Notably, we make the haptics function and will design the functions for sound and sensing.

3.3 System for Tracking Insects in Real-time

To implement the functions executed by cockroaches, such as object transportation, we proposed a system that can control and track their movement and send them signals for executing tasks. Notably, the proposed system can be implemented in real-time because of the possibility of real-time tracking and communication with the cockroach's backpack. As shown in Figure 2, we track the movement of the cockroaches in real-time based on the position of an affixed AR marker using a camera. Notably, based on our observations, a common method in the literature [58] might be incapable of identifying individuals in real-time. Second, to send the commands from the user's device, we utilized the communication mechanism using a radio-based station. Specifically, we set a radio-based station that communicated with each cockroach by sending the commands via a radio station to the circuit attached to each cockroach. Notably, we put the required items, including an AR marker and a circuit module, which we refer to as a backpack, on each cockroach.

3.4 Controlling Multiple Insects Reaching Their Respective Goals

In Figure 3, we propose a method for controlling multiple cockroaches towards their goals.

3.4.1 Set Goals with Shortest Paths. In this phase, goals are set with shortest paths as shown in Figure 3 (a). Given multiple goals of number n and cockroaches, we assign a goal for each cockroach without overlapping. Therefore, in all possible assignments of pairs of each goal and cockroach, we will take the assignment $\{(g_i, c_i)\}$, where each goal g_i is assigned with cockroach c_i and the total distance $\sum_{i=1}^n d_{c_i, g_i}$ is minimal among the total distance of each other assignment. Note that we utilize the AR markers attached to the cockroaches and the goals for computing the locations of cockroaches and goals and the distances among them. The coordinates of the cockroaches is updated every 500 milliseconds based on [72].

3.4.2 Controlling the Cockroaches to Attain Goals. In this phase, cockroaches are controlled for reaching each goals as shown in Figure 3 (b). We control the cockroaches to advance toward the goals based on the angles of the cockroaches toward their goals, their distances, and whether they are moving or not. Note that the angle is computed in a straight trajectory from the controlled cockroach to the goal based on the default horizontal line. When the angle is more than 10° or less than -10° , we make the cockroach move to the right or left, respectively, via electrically stimulating the left or right antenna automatically. Whenever the cockroach stop moving, the cerci are electrically stimulated to make it move forward. If the distance between the goal center and the cockroach is less than 10 cm, the stimulation to the cerci stop. This is to prevent the cockroaches from running out and stepping out of the goal area.

3.4.3 Keeping Cockroach at the Goal Area. In this phase, cockroaches are kept to stay at the goal area as shown in Figure 3 (c)-(f). For scenarios where a cockroach approaches the goal area, we propose control to keep the cockroach there as long as possible. We

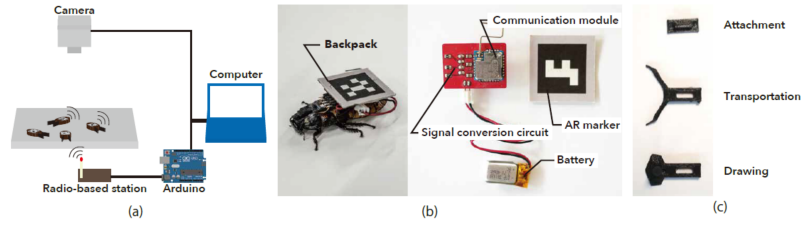


Figure 2: An on-hand, efficient, and scalable system of controlling multiple insects and their functions with AR markers, radio-based station, and flexible optional parts for functions.

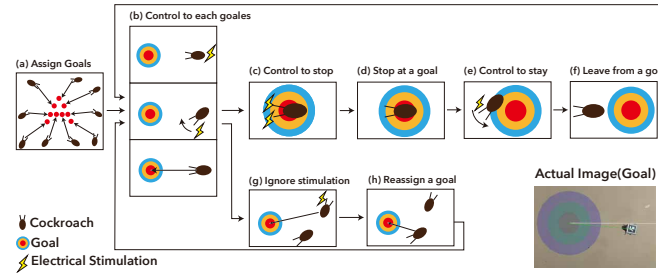


Figure 3: Multiple insects are controlled simultaneously for reaching their respective goals.

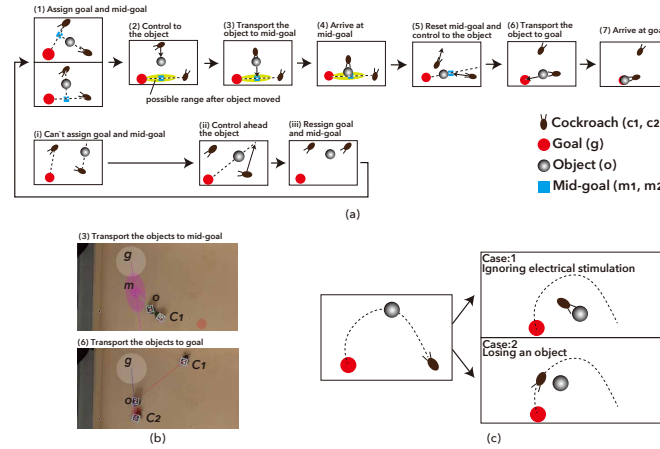


Figure 4: Multiple insects are controlled in cooperating to move an object to specified destinations. (b) Plan routes of transporting an object. (c) Difficulty of transporting an object by single insect in a curve route.

consider the distances between a cockroach and the goal center for 3 distances, 1 cm, 5 cm, and 10 cm. When the cockroaches are within 1 cm of the goal center, they are stimulated electrically. When they are between 1 to 5 cm away, we stop them electrically by stimulating both antennae continuously. When they are far from the goal center, between 5–10 cm away, we attempt to force them to return to the goal using a turnaround move. To perform the turnaround move, we control the cockroach to move either left or right, depending on which side it is facing.

3.4.4 Reassigning the Goals and Providing Substituted Cockroach Robots. In this phase, cockroaches are reassigned the goals as shown in Figure 3 (g) and (h). When controlling multiple cockroaches, some may become uncontrollable or desensitized to the electrical stimulus over time. Therefore, to increase the number of cockroaches reaching their goals, we propose reassigning goals to those that are not assigned goals. Note that a cockroach will be deemed unresponsive if it does not respond to stimulus for 100 frames (approximately 3.3s). In the mechanism, a cockroach is deemed unavailable (unassigned goals), if it is unresponsive and

far from the available goals. Further, we reassign goals to the available cockroaches at a specific time. Therefore, we may enhance the sustainability of a group of cockroach robots by running our system using available (nonworking) cockroaches and developing a mechanism to switch a robot between working and non-working status.

3.5 Object Transportation using Multiple Insects

We propose steps for transporting objects effectively using multiple cockroaches, as shown in Figure 4. Note that it might not be efficient to transport an object using only one cockroach, especially when it is necessary to move it through a winding path. For instance, in the scenario depicted in Figure 4 (c), it is easy to lose control over a cockroach when making turns. Therefore, multiple cockroaches are set up to cooperate to move an object. Each of them is positioned such that they move the object in straight lines. Hence, we may increase the accuracy of transporting objects by using multiple rather than single cockroaches and reduce the continuous controlling time on single individuals (when a cockroach completes the subtask of reaching a goal or transporting an object, it can attain nonactive status).

3.5.1 Select Short and Straight Paths. Given an object o , a goal g , and two cockroaches c_1 and c_2 , to make both cockroaches move in straight trajectories and shorten the total distance of paths, we propose two possible cases. In case one, cockroach c_1 moves the object to the mid-goal m_1 as a straight path p_{c_1, m_1} and thus cockroach c_2 moves the object to g through m_1 as the path $p_{c_2, g}$. In case two, inversely, cockroach c_2 moves the object to another mid-goal m_2 as a straight path p_{c_2, m_2} and thus cockroach c_1 moves the object to g through m_2 as the path $p_{c_1, g}$. We then set the paths, from either $p_{c_1, m_1} + p_{c_2, g}$ or $p_{c_2, m_2} + p_{c_1, g}$, which has a shorter distance for the two cockroaches.

3.5.2 One Cockroach Moves Object to the Mid-goal and the other Continues to Move it to Goal Destination. Based on the defined paths, the first cockroach c_a (either be c_1 or c_2) is controlled to move object o to the mid-goal; another cockroach c_b is kept on standby. When object o is moved to the mid-goal, cockroach c_b is controlled toward the location of o and move o to goal g using the reset straight paths. Note that the first cockroach c_a is prevented from disturbing cockroach c_b 's task by being controlled to move in other directions. Finally, when cockroach c_b moving o is at goal g (within 5 cm), c_b is controlled to leave g . In some cases of the relative positions among object o , goal g , and cockroaches, setting all the cockroaches to follow straight paths is not feasible and the need arises to move one cockroach to another location, at which the mid-goal can be set on the intersection of the line between a cockroach and object o and the line between another cockroach and goal g .

3.6 Optional Parts for Creation

We made the cockroach capable of executing given functions, such as object transportation, by affixing the necessary optional parts. Therefore, based on the function, we designed the shape, surface, and materials of the optional parts, which may be attached to the

external objects and touch the ground. In this study, we designed the optional parts for object transportation and the drawing and attachment required for the replacement optional parts.

3.6.1 Object Transportation. We designed the optional parts for the cockroach robots transporting objects while taking cognizance of cockroaches' natural tendency to avoid objects touching their antennae. Therefore, as shown in Figure 2 (c), we produced an optional part, a Y-shaped horn, for object transportation. The Y-shaped horn could be used to move objects by making the cockroach move forward in front of the objects with the given stimulus to both cerci. The design of the Y-shaped optional part was as follows. First, we had to prevent the cockroach from climbing on the object to be moved by adjusting the position of the installed optional part. Therefore, we avoided cockroaches from climbing on the object by setting a slope on the tip of the optional parts so that the optional parts would go underneath the object. The Y-shape horn can prevent the cockroach from losing objects, especially large ones, when it is moving from side to side (moving forward, left, or right). However, a large optional part may impede the mobility of the cockroach. Therefore, to investigate the cockroach's ability to move stably and with mobility, we set the size and weight of the optional part as 3 x 3 cm and 1 g, respectively.

3.6.2 Drawing. In addition, we designed the optional part for drawing (e.g., letters). In the drawing experiment using the controlled cockroach with a pen attached, we observed that the cockroach might find it hard to draw using a ballpoint pen, which may draw small, blurry lines. Therefore, we opted to attach a thick brush pen that could draw thick and legible lines when the cockroach is moving. In addition, we used a brush pen because the timing and pressure of the attached brush touching the ground were suitable.

4 USER STUDY: EXPECTED ON APPLICATIONS, PERFORMANCE, OR APPEARANCES IN DAILY LIFE CREATION TASKS

To analyze the practicality of co-creating artworks through the tool of insects, we conducted user studies on investigating user perception on the tool of cockroaches, user experiences on the functions of Calmbots, and their opinions and expectations about interesting tasks or novel functions of Calmbots. As shown in Table 1, there were nine participants of 8 males and 1 female (the average age was 22.11 years old), denoted by u1 - u9. Before user study, each participant was asked about the interest of cockroaches. At first, we showed each participant the Calmbots' functions of transportation, drawing, and haptics. After demo, each participant was asked about the five questions in Table 1.

We analyzed the feedback of nine participants. Most of users showed positive, convenient, or usable on the functions, especially on object transportation. Additionally, the mobility of cockroach robots was impressive. Further, users showed positive about Calmbots, although some of users were not interested in cockroaches. Users suggested many interesting or usable usages in living spaces. Nevertheless, some users expected the performance (e.g., control accuracies), appearance, and more applications.

Table 1: User study of participants about usage experiences with Calmbots.

<i>Before user study, Q: How do you feel about cockroaches?</i> Neutral: 3 users, Dislike: 6 users
<i>Q1. What is your feedback about the demo of object transportation?</i> Interested: 8 users of 9; Convenient: 7 users of 9 Convenient to carry various things. (u1, u9) Useful for humans. (u2, u5) Excellent mobility. (u4, u6) Interesting concept. (u7)
<i>Q2. What is your feedback about the demo of drawing?</i> Interested: 7 users; Convenient: 3 users Artistic, different with human-written characters. (u8, u9)
<i>Q3. What is your feedback about the demo of haptics?</i> Interested: 6 users; Convenient: 3 users Treated as pet or plant. (u1, u2) Dislike to touch cockroaches. (u5, u6) Convenient than a wearable device. (u4) Non wearable and not disturbing at-hand work. (u8)
<i>Q4. Which one of the above three functions would you like to use?</i> Object transportation: 8 users, Drawing: 1 user
<i>Q5. In which daily life situations, would you consider to use Calmbots (cockroach robots)? Or what kind of novel functions would be interesting?</i> Cleaning. (u1, u5, u6, u8, u9) Concerned with accuracy on haptics or drawing. (u1, u2, u4, u5) Process things at place people is difficult to reach. (u2, u8, u9) Remote control or switch for device. (u4, u6) As memo note. (u4, u5) Wake up people when emergency. (u7, u9) As movable or flying monitor. (u8, u9) Paint on the wall, such as Graffiti. (u5, u6) As toy, adornment, or pet. (u1, u3) To improve appearance of cockroach robots. (u4, u7) Changing haptics using novel attachment is interesting. (u3)

5 DISCUSSION AND CONCLUSIONS

5.1 Analysis of Experimental Results

5.1.1 The Controllability of Madagascar Cockroaches was Considered. The experimental results proved that the proposed method was effective for controlling multiple cockroaches to reach defined goals (the reaching accuracy was quality). However, the following issues were observed. When multiple cockroaches were controlled for moving objects to defined points, some of them collided (e.g., climbing on top of another), and as a result, their backpacks could get tangled up, which could make the AR markers temporarily untraceable. In controlling multiple cockroaches toward their goals, individuals that have already attained their goals within the reach of others could become less controllable. Additionally, the reaching time of an individual could not be estimated accurately. Therefore, in achieving multiple goals using the cockroaches, the entire performance may be improved by providing more substitute cockroaches for the tasks, that is, deploying as many cockroaches as possible for a task (goal).

5.1.2 Appropriate Tasks Conducted by Cockroach Robots. We observed that controlling cockroaches such that they remained in

specific areas was difficult. Therefore, *Calmbots* might not be effective for tasks that require responsiveness or cost-intensive tasks. This implies that tasks that do not require responsiveness, can be executed off-line, or at a delayed time can be performed using cockroach robots. *Calmbots* may be suitable for moving nonprecious objects or executing the carriage of objects at a later specified time or when the users are not present. In addition, we observed that the controlling accuracy for the cockroaches could decrease over time and the cockroaches could be affected by internal or external factors. Therefore, we can design a system that can sort the cockroaches according to usability and control time and, based on this, assigned tasks at suitable timing.

5.1.3 Effect of Backpack on Cockroaches. We discussed the effect of backpacks on the mobility of cockroaches. Note that a cockroach's mobility and balance are related. We observed that when the backpacks were off-balance, sometimes the cockroaches were unable to crawl on vertical surfaces and took longer crawling over obstacles (e.g., cables). Additionally, when multiple cockroaches were close to each other, their accessories (e.g., antennae) could get tangled, which reduced their mobility. Therefore, to reduce the effect of the backpacks, we propose putting the backpacks close to the center of

gravity of the cockroach's body and reducing the backpack weight by using lightweight materials.

5.1.4 Feasibility of Calmbots as Interfaces. We discussed the performance and feasibility of *Calmbots* as interfaces in living spaces. Madagascar hissing cockroaches are silent and self-sustaining; furthermore, they have high mobility. Based on the experiments, we implemented a few function (drawing, haptics, and display) to *Calmbots* as shown in Figure 1. However, there are some concerns about the control accuracy for multiple cockroaches. In use as a display (Figure 1), we may need greater control accuracy because the display requires multiple cockroaches to occupy specific locations (long staying time may be needed). In addition, when using them as pixels, the proper directions they face are required since the shape of each cockroach is oval. For transportation or haptics in living spaces, *Calmbots* may be feasible because they have demonstrable reaching ability and considerable staying time. However, for haptics proper tactile should be presented by devising optional parts in terms of *calm computing*. Furthermore, although cockroaches can be used to draw large characters, drawing small characters is a challenge owing to the size of a cockroach and requisite control accuracy. Within a certain workload of continuous controlling time, the system may perform short-period tasks in living spaces. From the current experimental performance, tasks requiring high responsiveness and precise control may not be suitable for *Calmbots*. Therefore, the tasks requiring a long duration or that can be repeatedly executed may be appropriate for *Calmbots*.

5.2 Realizing Flexible Insects' Interfaces towards Co-creation

5.2.1 Interfaces of Hybrid Cockroaches and Robots. We observed limitations in the functionality of the cockroaches, such as the control performance decreasing as the number of cockroaches increased. To deploy *Calmbots* in living spaces, we hold that novel interfaces integrating cockroaches and common robots may have the complementary advantages of cockroaches and robots. In terms of mobility, common robots can move easily under ideal conditions, such as flat or dry floors; cockroaches, on the other hand, are movable under diverse conditions, such as narrow paths or vertical floors. Additionally, the entire control performance can be improved by using cockroaches, owing to their ability to execute diverse tasks in living spaces, such as tasks needing quiet or minimal noise.

5.2.2 Flexible Optional Parts. The optional parts can be customized for realizing multiple operations such as object transportation and drawing. In object transportation, to move a large object, we may choose to use optional parts with suitable shapes or a high shape-tack to attach the large object. Similarly, for drawing using a specified pen or representing specified expressions, we may use an optional part that can attach the pen and set it in the right position for the expression. In addition, users can customize the design of the optional parts. Note that optional parts can be flexibly customized using digital fabrication, such as 3D printing or computer-aided design.

5.2.3 Factors Affecting Insect Robots used in Living Spaces. We examined other factors such as biological behaviors or environmental effect. For example, cockroaches may have specific behaviors pertaining to food intake or excretion, which can necessitate the presence of additional things, such as food, around the circuits or backpacks, thus possibly triggering a short-circuit. Additionally, co-existing creatures (e.g., cat) might induce unusual behaviors in the cockroach robots. For such contexts, we may propose ideas that are unrelated to the (*Calmbots*) system, such as limiting the movable area of the cockroach robots. We can consider another control method that is not based on electric stimulation such as utilizing special materials or spices. For instance, we could use Vaseline petroleum jelly to reduce the movement of the cockroaches. Note that there are other problems pertaining to users or appropriate usage situations.

5.3 Co-creation Capabilities, Feasibilities, and Expectations with Calmbots

Towards creation using tools of insects, we proposed *Calmbots* with functions of drawing, display, transportation, and haptics, and option parts. Through experimental analysis, the performance (controllability and mobility of multiple cockroaches) of *Calmbots* on several daily-life tasks might be effective and the sustainability was evaluated under certain usage duration of minutes. In interviews, most users showed positive feedbacks on transportation, drawing, or haptics, and expected multiple interesting or usable tasks in living spaces or remote conditions, such as remote painting on the wall as graffiti, remote sensing or object transportation at place people hard to reach, and dynamic haptics with high mobility. On the other hand, most users showed interested but might concern about some issues on performance, usability, appearance of *Calmbots*. E.g., some users preferred remote control of *Calmbots* rather than applying functions in close range, which may be due to user's feeling or appearance of *Calmbots*. Some users concerned the quality of *Calmbots* performed on tasks required high details of contents. From above analysis, considering co-creation activities using *Calmbots*, we might infer as follows. Some users might be interested in conducting simple creation or artworks, which can be indirectly co-created through capabilities of *Calmbots*, or those artworks can be physically distant with artist users. That is, users would like to co-create simple and rough artworks indirectly or remotely through tool of high movable and controllable insects. Further, not only artworks and artist users can be distant, to increase quality of artworks, attachable option parts can be used to increase capability or creativity of insect-artists. Suitable creation activities with current *Calmbots* might be remote ways of rough painting, printmaking, sculpture, or drawing on or with simple objects, such as objects with rough shapes (rectangles, curves, spiral, or diagonals), or objects with materials or properties easy to be modified.

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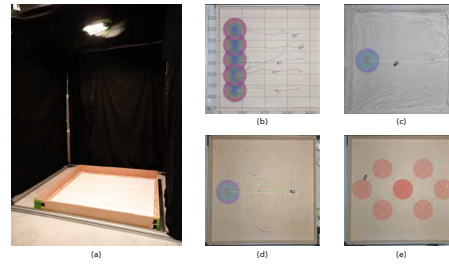
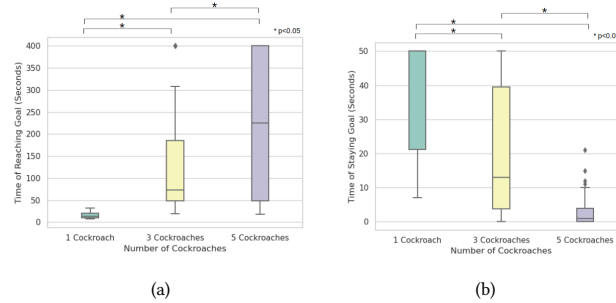
A EXPERIMENTAL RESULTS

A.1 Controlling Performance of Multiple Cockroaches on Reaching Goals

We evaluated the success of controlling multiple cockroaches simultaneously to reach the defined goals. In our method, the factors were the criterion of electrical stimulus, tracking or communicating with cockroach backpacks using AR markers and a radio station, and the proposed method with multiple parameters (e.g., the angle differences between the goals and cockroaches). In each experiment, each cockroach was aligned horizontally and controlled to reach the goal, which was 70-cm away in the same direction. The time

Table 2: Calmbots and related robots works.

	<i>Zooids</i> [19]	<i>HERMIT</i> [44]	<i>TUI (inFORM)</i> [17]	<i>Swarmrobots</i> (Rolling Pixels, shaped bots) [29]	<i>Calmbots (by Cockroaches)</i>
Applicable range	Flat floors	Uneven floors	Limited range	Flat floors	Ubiquitous
Robot scale	cm	cm	cm	cm	cm
Maintenance (energy)	Electrical	Electrical	Electrical	Electrical	Biological (themselves)
Silent (when using)	x (Motor)	x (Motor)	x (Pin actuators)	x (Motor)	o (Muscle)
Control accuracy	o	o	o	o	Δ (Effective reaching, staying was limited)
Controllable time	o	o	o	o	Δ (Controlled under certain duration)

**Figure 5: Experiment setup: (a) closed room without shadow for conducting experiments, and the experiments for (b) controlling multiple cockroaches on reaching, (c) mobility on a floor of carpet, (d) mobility on a floor with cables, and (e) workload on figure-eight routes.****Figure 6: Controlling performance of multiple cockroaches on reaching goals in (a) reaching times and (b) staying times.**

taken to reach the goal area and the duration of remaining within it was measured for each cockroach, as shown in Figure 5 (b). We experimented on three cockroach teams of varying numbers, one cockroach, three cockroaches (with one for substitution), and five cockroaches (with one for substitution), to observe the effect of the number of cockroaches on the control performance. We included multiple cockroaches in each team to ensure the fairness of the performance evaluation. In the experiment, we used four teams, each with unique individuals. We had 40 trials, 10 for each team.

A.1.1 Results. As shown in Figure 6 (a), the time taken to perform a task by one cockroach (the median time: 13.5s) was significantly better than that of three or five cockroaches. Additionally, the

variances of the reaching times of 3 or 5 cockroaches were high, such that the lower to the upper quartiles were between 50 s to 200 s. Similarly, for the staying times in Figure 6 (b), 1 cockroach could be controlled to stay for approximately 21 s, whereas 3 and 5 cockroaches had less staying time, less than 5 s in many instances. These results implied that the performance of our control method decreased when more cockroaches were being controlled to reach their goals.

A.2 Effect of Floor Flatness on Mobility

We evaluated the mobility of cockroaches based on the floor flatness. Specifically, we performed experiments on uneven floors, such as

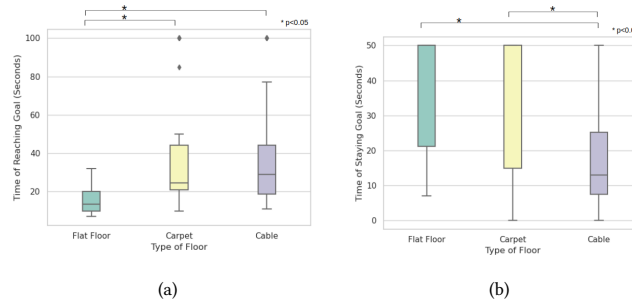


Figure 7: Mobility performance of cockroach individuals on the affect of floor flatness, in (a) reaching times and (b) staying times.

those with cables, on which typical robots Toio (Sony)² cannot move. We evaluated the effect of the flatness of floors, such as a flat, carpeted floor (Fig. 5 (c)), and cable-lined floor (Fig. 5 (d)), on the mobility of a cockroach approaching a destination 70 cm away. We used 4 individual cockroaches for 10 trials each, making a total of 40 trials.

A.2.1 Results. As shown in Figure 7 (a), the arrival time of one cockroach on a flat floor was significantly better than on the carpeted and cable-lined floors. Further, as shown in Figure 7 (b), one cockroach could be controlled to stay on a flat floor or carpet, as opposed to the cable-lined floor. These results might imply the following. First, we showed that, unlike typical robots, cockroaches could be controlled to move on an uneven floor. Note that one cockroach was more controllable on flat floors than on uneven floors, such as carpeted or cable-lined floors. Further, we observed that a cockroach could be controlled to stay longer on an uneven floor such as carpet, even if the control over it had diminished and mobility was impeded. This might be because the movement of cockroaches may be impeded by the materials of uneven floors (carpet). Similarly, on the cable-lined floors, cockroaches might have low mobility because the backpacks on them might get caught, thus affecting the movement of the cockroach; for instance, the center of gravity of the cockroach's body could be shifted backward.

A.3 Analysis of Continuous Control Success on Individual Cockroaches and Stimulated Body Parts

We evaluated the workload of the system when individual cockroaches were being continuously controlled. Note that a cockroach under continuous electrical stimulation can become desensitized, and consequently, uncontrollable. In each experiment, a cockroach was continuously controlled to move on a figure-eight route recursively until it could no longer be controlled, as shown in Figure 5 (e). Further, we evaluated the effectiveness of stimulating each of three body parts, i.e., left antenna, right antenna, and cerci. Note that a cockroach was deemed uncontrollable if it remained unresponsive even after repeated stimulation. We used 4 individuals and

performed 10 trials for each of them. Further, we evaluated the cockroaches based on their usability over a long duration comprising working and nonworking conditions. Therefore, we analyzed the controlling performance of the same individuals or body parts during consecutive experiments with a time-break of 20 min between each.

A.3.1 Results for Individual Cockroaches. In Figure 8 (a), among the results for 4 individuals, we observed that the medians of continuous controlling times for 3 individuals (Cockroach ID 1, ID 3, and ID 4) were more than 167.5 s, and the variances of the controlling times of 2 individuals (Cockroach ID 1 and ID 4) were large. We can infer that individual cockroaches can be continuously controlled over a period of time. However, this may result in instability.

A.3.2 Results for Body Parts and Last Part of Route. In Figure 8 (b), among the results for three body parts, we observed that the times for the left and right antennae were close, and their values might be bigger than those for the cerci. Note that the route was a figure eight. We observed that on this route, with more turns than straight lines, controlling cockroaches required more stimulations to the left or right antennae, which corresponded to the left or right directions, compared to the cerci for straight direction. Additionally, a similar number of stimulations was required for both antennae, indicating that the route had equal left and right turns. The results might be considered as proof that our control methods for directions were effective and consistent with the expected movement of cockroaches. Finally, we observed the performance on the last part of a route where the cockroach's response to stimulation might have deteriorated. In Figure 8 (c), the left or right antenna required greater stimulations than the cerci. This might imply that, when the response to the stimuli had deteriorated, it was more useful to stimulate the antennae (for left or right directions) instead of the cerci (for straight direction).

B ETHICAL CONSIDERATION ABOUT HUMAN-INSECT EXPERIMENTS AND USER STUDIES

To make our experiments using insects and user studies sustainably and ethically, we have applied ethical review in our institution, received feedback and revised experiments and user studies, and

²Retrieved April 3, 2021, from <https://www.sony.com/en/SonyInfo/design/stories/toio/>

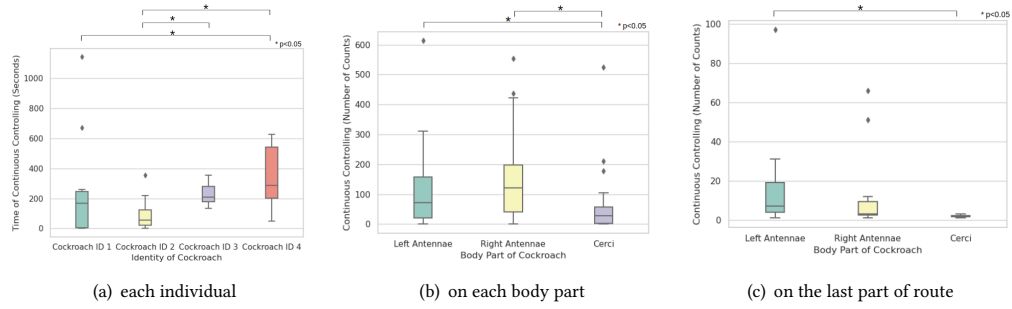


Figure 8: Performance of continuous controlling on cockroach individuals.

received admission of experiments and user studies. In experiments using cockroaches, we paid attention on providing well-living conditions during experiments and off-experiments for cockroaches and kept clean experimental environments. Note that we also followed the ethics, guidelines, maintenance of welfare in insect or animal studies [18, 66]. To make user studies comfortably and ethically, we detailed the experimental procedures to users in advance,

especially about the haptics or vision contact with the cockroach tool. In interview, we paid attention on conducting comfortable interview by avoiding sensitive or unwilling contents. Note that the duration or items of user study were adjusted or reduced according to the feedback or willingness of users on interaction with the cockroach tool.